# Russia — EDA, TTC-модели, SHAP

**Что тут:** EDA → Stage-1 (Through-The-Cycle) → проверка H1 и H2 по России.

- **H1.** Ансамбли (RF, XGBoost) выдают более высокий ROC-AUC, чем Logit, но у них больше **зазор train–test** (переобучение).
- **H2.** Сумма `|SHAP|` по блокам **Ликвидность + НМА** выше, чем по блоку **Леверидж**.

**Правила научрука:** без SMOTE, `class_weight` ≈ 1:10, stratified split.

**Артефакты** сохраняются в `reports/russia/` — все графики как PNG, таблицы как CSV, финальные вердикты в `ru_summary.md`. Этого достаточно, чтобы вставить в текст диплома.

In [1]:
```
import warnings
warnings.filterwarnings('ignore')

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from pathlib import Path

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.metrics import roc_auc_score, average_precision_score

from xgboost import XGBClassifier
import shap

RNG = 42
sns.set_theme(style='whitegrid', context='notebook')
plt.rcParams['figure.dpi'] = 110

PROCESSED = Path('../data/processed')
REPORTS = Path('../reports/russia')
REPORTS.mkdir(parents=True, exist_ok=True)

def save_fig(fig, name):
    path = REPORTS / f'{name}.png'
    fig.savefig(path, dpi=160, bbox_inches='tight')
    print(f'  saved → {path}')

# Сюда копим текстовые выводы для итогового markdown-отчёта
_report_lines = ['# Russia — Stage-1 (TTC) results\n']
def log(msg=''):
    print(msg)
    _report_lines.append(msg)

df = pd.read_csv(PROCESSED / 'ru_panel_cleaned.csv', encoding='utf-8-sig')
```

```
print('Shape:', df.shape)
df.head(3)
```

```
Shape: (88714, 20)
```

Out[1]:

| | Регистрационный номер | year | EBIT | Активы всего | Внеоборотные активы | Выручка | Д задо |
|---|---|---|---|---|---|---|---|
| **0** | 1022401483928 | 2012 | 4294000.0 | 10581000.0 | 6784000.0 | 3744000.0 | |
| **1** | 1022401483928 | 2013 | 2999000.0 | 12968000.0 | 6530000.0 | 3906000.0 | |
| **2** | 1022401483928 | 2014 | 2614000.0 | 18961000.0 | 17531000.0 | 4111000.0 | |

# 1. Feature engineering

Формируем ratios по блокам H2:

- **Liquidity:** `current_ratio`, `cash_to_assets`, `cash_to_cl`, `wc_to_assets`
- **Innovation:** `intangibles_to_assets`
- **Leverage:** `debt_to_assets`, `debt_to_equity`, `lt_debt_to_assets`, `interest_coverage`
- **Profitability:** `roa`, `net_margin`, `operating_margin`, `cfo_to_assets`
- **Size:** `log_assets`, `log_revenue`

Винзоризация 1/99% — типичные IT-ратио сильно скошены.

In [2]:

```
ID_COL = 'Регистрационный номер'
TARGET = 'is_bankrupt'

A  = 'Активы  всего'
CA = 'Оборотные активы'
CASH = 'Денежные средства и денежные эквиваленты'
INT_A = 'Нематериальные активы'
EQ = 'Капитал и резервы'
LT_L = 'Долгосрочные обязательства'
ST_L = 'Краткосрочные обязательства'
REV = 'Выручка'
EBIT = 'EBIT'
NI = 'Чистая прибыль (убыток)'
INTEREST = 'Проценты к уплате'
CFO = 'Сальдо денежных потоков от текущих операций'

def safe_div(a, b):
    b = b.replace(0, np.nan)
    return a / b

panel = df.copy()
total_debt = panel[LT_L] + panel[ST_L]

panel['current_ratio']         = safe_div(panel[CA], panel[ST_L])
panel['cash_to_assets']        = safe_div(panel[CASH], panel[A])
panel['cash_to_cl']            = safe_div(panel[CASH], panel[ST_L])
```

```
panel['wc_to_assets']          = safe_div(panel[CA] - panel[ST_L], panel[A])

panel['intangibles_to_assets'] = safe_div(panel[INT_A], panel[A])

panel['debt_to_assets']        = safe_div(total_debt, panel[A])
panel['debt_to_equity']        = safe_div(total_debt, panel[EQ])
panel['lt_debt_to_assets']     = safe_div(panel[LT_L], panel[A])
panel['interest_coverage']     = safe_div(panel[EBIT], panel[INTEREST])

panel['roa']              = safe_div(panel[NI], panel[A])
panel['net_margin']       = safe_div(panel[NI], panel[REV])
panel['operating_margin'] = safe_div(panel[EBIT], panel[REV])
panel['cfo_to_assets']    = safe_div(panel[CFO], panel[A])

panel['log_assets']  = np.log1p(panel[A].clip(lower=0))
panel['log_revenue'] = np.log1p(panel[REV].clip(lower=0))

FEATURES = [
    'current_ratio', 'cash_to_assets', 'cash_to_cl', 'wc_to_assets',
    'intangibles_to_assets',
    'debt_to_assets', 'debt_to_equity', 'lt_debt_to_assets', 'interest_coverage'
    'roa', 'net_margin', 'operating_margin', 'cfo_to_assets',
    'log_assets', 'log_revenue',
]

FEATURE_GROUPS = {
    'Liquidity':     ['current_ratio', 'cash_to_assets', 'cash_to_cl', 'wc_to_as
    'Innovation':    ['intangibles_to_assets'],
    'Leverage':      ['debt_to_assets', 'debt_to_equity', 'lt_debt_to_assets', '
    'Profitability': ['roa', 'net_margin', 'operating_margin', 'cfo_to_assets'],
    'Size':          ['log_assets', 'log_revenue'],
}

panel[FEATURES] = panel[FEATURES].replace([np.inf, -np.inf], np.nan)
for c in FEATURES:
    lo, hi = panel[c].quantile([0.01, 0.99])
    panel[c] = panel[c].clip(lo, hi)
panel[FEATURES] = panel[FEATURES].fillna(panel[FEATURES].median(numeric_only=Tru

panel[FEATURES + [TARGET]].describe().T.round(3).to_csv(REPORTS / 'ru_feature_st
print(f'Фичи: {len(FEATURES)}  |  stats → reports/russia/ru_feature_stats.csv')
panel[FEATURES].describe().T.round(3)
```

```
Фичи: 15  |  stats → reports/russia/ru_feature_stats.csv
```

Out[2]:

| | count | mean | std | min | 25% | 50% | 75% | |
|---|---|---|---|---|---|---|---|---|
| **current_ratio** | 88714.0 | 10.737 | 29.845 | 0.012 | 1.117 | 2.269 | 6.739 | 22 |
| **cash_to_assets** | 88714.0 | 0.287 | 0.312 | 0.000 | 0.030 | 0.155 | 0.476 | |
| **cash_to_cl** | 88714.0 | 3.398 | 10.731 | 0.000 | 0.067 | 0.402 | 1.673 | 8 |
| **wc_to_assets** | 88714.0 | -0.253 | 4.386 | -38.481 | 0.087 | 0.482 | 0.800 | |
| **intangibles_to_assets** | 88714.0 | 0.077 | 0.417 | 0.000 | 0.000 | 0.000 | 0.000 | |
| **debt_to_assets** | 88714.0 | 3.299 | 16.626 | 0.004 | 0.188 | 0.527 | 0.964 | 14 |
| **debt_to_equity** | 88714.0 | 9.197 | 46.819 | -42.986 | 0.072 | 0.455 | 2.171 | 37 |
| **lt_debt_to_assets** | 88714.0 | 1.045 | 5.713 | 0.000 | 0.000 | 0.000 | 0.122 | 5 |
| **interest_coverage** | 88714.0 | 165.278 | 1267.100 | -1460.333 | 9.355 | 9.355 | 9.355 | 1704 |
| **roa** | 88714.0 | 0.211 | 1.695 | -8.700 | 0.004 | 0.135 | 0.467 | 1 |
| **net_margin** | 88714.0 | -0.002 | 0.782 | -5.999 | 0.002 | 0.053 | 0.191 | |
| **operating_margin** | 88714.0 | 0.026 | 0.736 | -5.560 | 0.007 | 0.068 | 0.214 | |
| **cfo_to_assets** | 88714.0 | 0.037 | 0.553 | -2.694 | 0.000 | 0.000 | 0.000 | |
| **log_assets** | 88714.0 | 15.276 | 2.346 | 9.210 | 13.813 | 15.321 | 16.840 | 2 |
| **log_revenue** | 88714.0 | 16.178 | 1.975 | 10.597 | 14.902 | 16.159 | 17.498 | 2 |

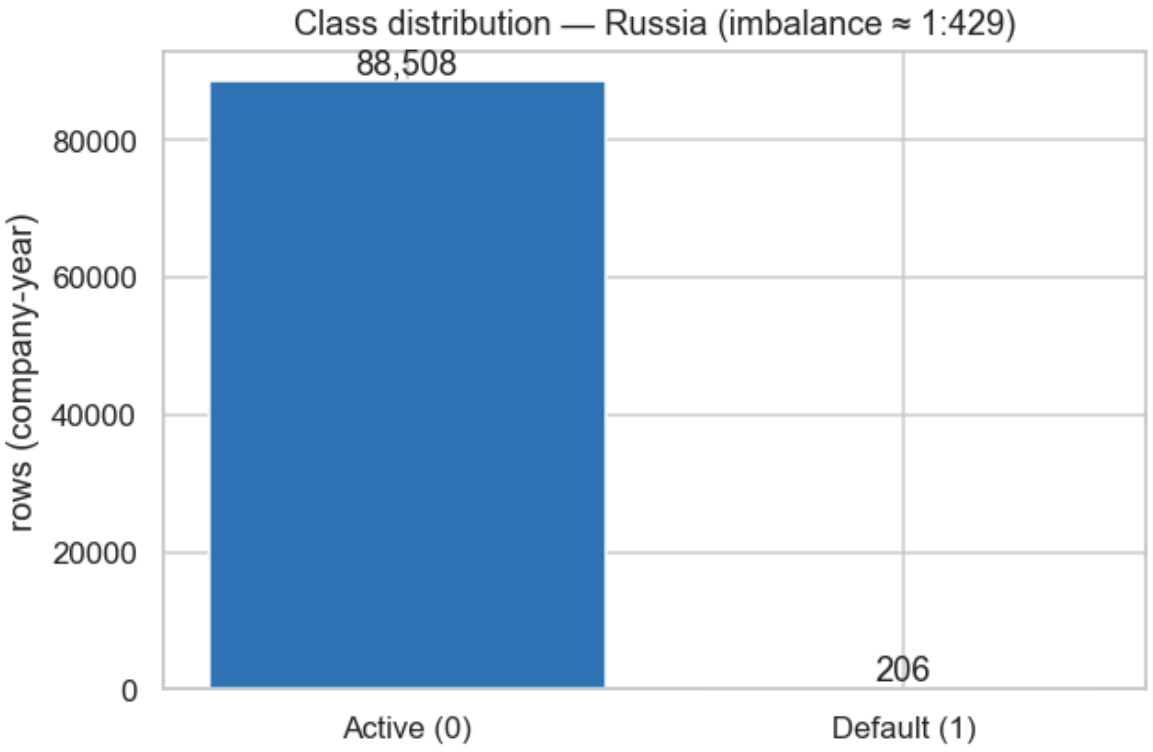
# 2. EDA

## 2.1 Классовый дисбаланс

In [3]:

```
vc = panel[TARGET].value_counts()
n_active  = int(vc.get(0, 0))
n_default = int(vc.get(1, 0))
ratio = n_default / (n_active + n_default)
counts = pd.Series({'Active (0)': n_active, 'Default (1)': n_default})

fig, ax = plt.subplots(figsize=(6, 4))
bars = ax.bar(counts.index, counts.values, color=['#2E75B6', '#C00000'])
ax.set_title(f'Class distribution — Russia (imbalance ≈ 1:{n_active//max(n_defau
ax.set_ylabel('rows (company-year)')
for b, v in zip(bars, counts.values):
    ax.text(b.get_x()+b.get_width()/2, v, f'{v:,}', ha='center', va='bottom')
plt.tight_layout()
save_fig(fig, '01_class_distribution'); plt.show()

log(f'\n## Class balance\n* Active (0): **{n_active:,}**')
log(f'* Default (1): **{n_default:,}**')
log(f'* Positive share: **{ratio*100:.3f}%**  (≈ 1:{n_active//max(n_default,1)})
```

```
saved → ..\reports\russia\01_class_distribution.png
```

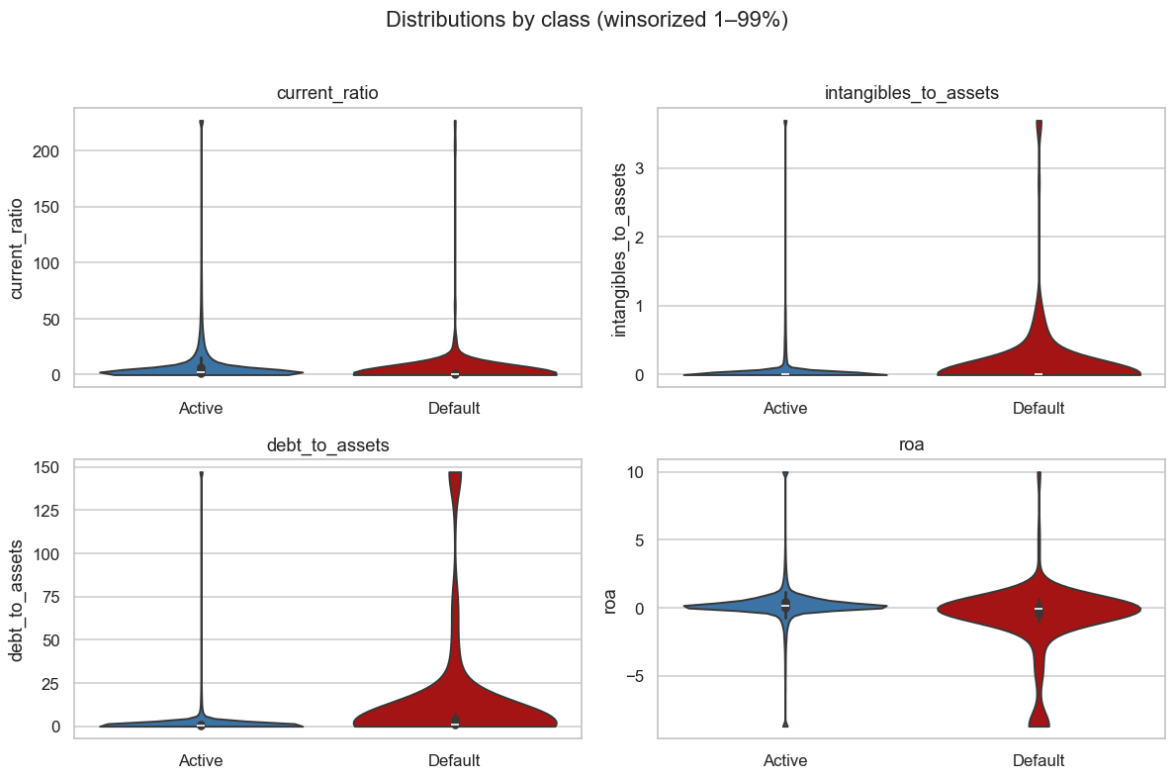



```
## Class balance
* Active (0): **88,508**
* Default (1): **206**
* Positive share: **0.232%**  (≈ 1:429)
```

## 2.2 Violin plots — по классу

In [4]:
```
panel['_lbl'] = panel[TARGET].map({0:'Active', 1:'Default'})
show = ['current_ratio', 'intangibles_to_assets', 'debt_to_assets', 'roa']

fig, axes = plt.subplots(2, 2, figsize=(11, 7))
for ax, col in zip(axes.ravel(), show):
    sns.violinplot(data=panel, x='_lbl', y=col, ax=ax,
                   palette={'Active':'#2E75B6','Default':'#C00000'}, cut=0)
    ax.set_title(col); ax.set_xlabel('')
plt.suptitle('Distributions by class (winsorized 1–99%)', y=1.02)
plt.tight_layout()
save_fig(fig, '02_violin_by_class'); plt.show()

med = panel.groupby('_lbl')[show].median().round(3)
med.to_csv(REPORTS / 'ru_medians_by_class.csv', encoding='utf-8-sig')
med
```

```
saved → ..\reports\russia\02_violin_by_class.png
```




Out[4]:

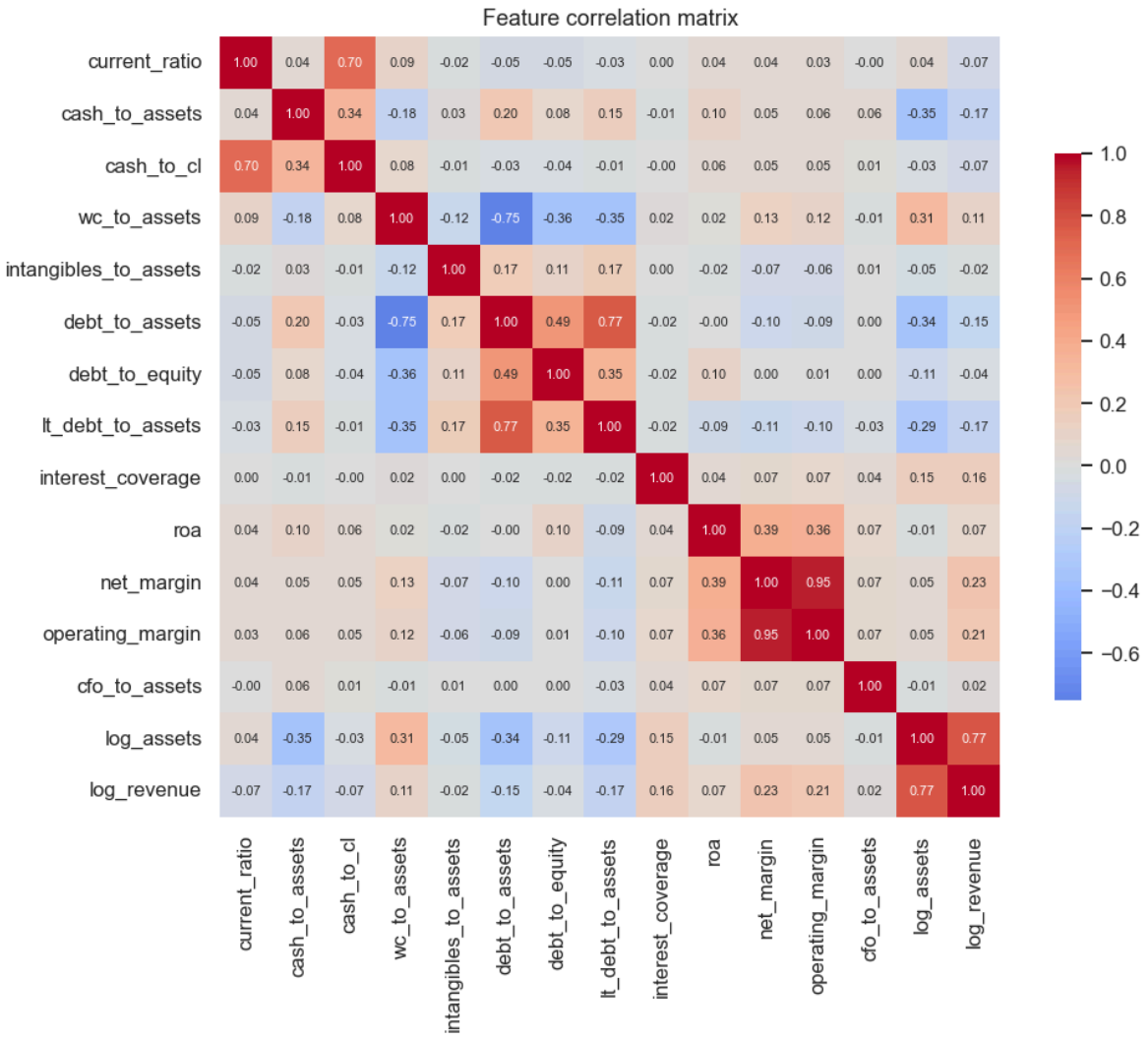
| _lbl | current_ratio | intangibles_to_assets | debt_to_assets | roa |
|---|---|---|---|---|
| **Active** | 2.276 | 0.0 | 0.526 | 0.135 |
| **Default** | 0.842 | 0.0 | 1.124 | -0.031 |

## 2.3 Корреляционная матрица

In [5]:

```
corr = panel[FEATURES].corr()
fig, ax = plt.subplots(figsize=(10, 8))
sns.heatmap(corr, cmap='coolwarm', center=0, annot=True, fmt='.2f',
            square=True, cbar_kws={'shrink':0.7}, annot_kws={'size':7}, ax=ax)
ax.set_title('Feature correlation matrix')
plt.tight_layout()
save_fig(fig, '03_correlation_heatmap'); plt.show()

corr.round(3).to_csv(REPORTS / 'ru_correlation_matrix.csv', encoding='utf-8-sig'

high = (corr.abs() > 0.85) & (corr.abs() < 1.0)
pairs = [(i,j,corr.loc[i,j]) for i in corr.index for j in corr.columns if i<j an
print('|corr| > 0.85:')
for p in pairs: print(f'  {p[0]} × {p[1]}: {p[2]:+.2f}')
if not pairs: print('  нет — мультиколлинеарность под контролем')
```

```
saved → ..\reports\russia\03_correlation_heatmap.png
```




```
|corr| > 0.85:
  net_margin × operating_margin: +0.95
```

# 3. Train / Test split

Stratified 80/20. Temporal split отложен до Stage-2 (PIT).

In [6]:
```
X = panel[FEATURES].values
y = panel[TARGET].values

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, stratify=y, random_state=RNG
)
print(f'Train: {X_train.shape},  defaults = {y_train.sum()}  ({y_train.mean()*10
print(f'Test:  {X_test.shape},  defaults = {y_test.sum()}  ({y_test.mean()*100:.
```

```
Train: (70971, 15),  defaults = 165  (0.23%)
Test:  (17743, 15),  defaults = 41  (0.23%)
```

# 4. Stage-1 модели (TTC)

`class_weight={0:1, 1:10}` (Logit/RF) и `scale_pos_weight=10` (XGBoost).

In [7]:
```
POS_WEIGHT = 10
```

```
logit = Pipeline([
    ('sc', StandardScaler()),
    ('clf', LogisticRegression(max_iter=2000, class_weight={0:1, 1:POS_WEIGHT},
                               solver='liblinear', random_state=RNG)),
])
rf = RandomForestClassifier(
    n_estimators=400, max_depth=None, min_samples_leaf=3,
    class_weight={0:1, 1:POS_WEIGHT}, n_jobs=-1, random_state=RNG,
)
xgb = XGBClassifier(
    n_estimators=500, max_depth=5, learning_rate=0.05,
    subsample=0.9, colsample_bytree=0.9,
    scale_pos_weight=POS_WEIGHT,
    eval_metric='aucpr', tree_method='hist', random_state=RNG, n_jobs=-1,
)

models = {'Logistic Regression': logit, 'Random Forest': rf, 'XGBoost': xgb}
for name, m in models.items():
    m.fit(X_train, y_train)
    print(f'✓ trained {name}')
```

```
✓ trained Logistic Regression
✓ trained Random Forest
✓ trained XGBoost
```

# 5. H1 — Train vs Test, зазор (overfit check)

In [8]:

```
def scores(model, X, y):
    if hasattr(model, 'predict_proba'):
        p = model.predict_proba(X)[:, 1]
    else:
        p = model.decision_function(X)
    return roc_auc_score(y, p), average_precision_score(y, p)

rows = []
for name, m in models.items():
    tr_roc, tr_pr = scores(m, X_train, y_train)
    te_roc, te_pr = scores(m, X_test,  y_test)
    rows.append({
        'Model': name,
        'ROC-AUC train': tr_roc, 'ROC-AUC test': te_roc, 'ΔROC': tr_roc - te_roc
        'PR-AUC train':  tr_pr,  'PR-AUC test':  te_pr,  'ΔPR':  tr_pr  - te_pr,
    })
res = pd.DataFrame(rows).set_index('Model').round(4)
res.to_csv(REPORTS / 'ru_h1_metrics.csv', encoding='utf-8-sig')
print('metrics → reports/russia/ru_h1_metrics.csv')
res
```

```
metrics → reports/russia/ru_h1_metrics.csv
```

Out[8]:

| Model | ROC-AUC train | ROC-AUC test | ΔROC | PR-AUC train | PR-AUC test | ΔPR |
|---|---|---|---|---|---|---|
| **Logistic Regression** | 0.8789 | 0.8548 | 0.0241 | 0.0581 | 0.1090 | -0.0509 |
| **Random Forest** | 0.9998 | 0.9016 | 0.0982 | 0.8846 | 0.0594 | 0.8253 |
| **XGBoost** | 0.9999 | 0.8814 | 0.1185 | 0.9583 | 0.0411 | 0.9172 |

In [9]:

```
# H1 verdict
best_test_roc = res['ROC-AUC test'].idxmax()
logit_delta = res.loc['Logistic Regression', 'ΔROC']
ens_delta_max = res.loc[['Random Forest','XGBoost'], 'ΔROC'].max()
ens_test_max  = res.loc[['Random Forest','XGBoost'], 'ROC-AUC test'].max()
logit_test    = res.loc['Logistic Regression', 'ROC-AUC test']

log('\n## H1 — overfitting на малом числе дефолтов')
log(f'* Best test ROC-AUC: **{best_test_roc}** ({res.loc[best_test_roc,"ROC-AUC
log(f'* Logit ΔROC (train-test): **{logit_delta:+.4f}**')
log(f'* Max ensemble ΔROC:        **{ens_delta_max:+.4f}**')

cond_a = ens_test_max > logit_test
cond_b = ens_delta_max > logit_delta

if cond_a and cond_b:
    verdict = '✅ **H1 ПОДТВЕРЖДАЕТСЯ**: ансамбли точнее на тесте, но переобучак
elif cond_a and not cond_b:
    verdict = '⚠️ H1 частично: ансамбли точнее, но Logit НЕ стабильнее (Δ сопост
elif not cond_a and cond_b:
    verdict = '⚠️ H1 частично: Logit сопоставим по точности, но ансамбли переобу
else:
    verdict = '❌ H1 НЕ подтверждается: ансамбли не дают прироста и не хуже по з
log(f'\n{verdict}')
print('\n' + verdict)
```

```
## H1 — overfitting на малом числе дефолтов
* Best test ROC-AUC: **Random Forest** (0.9016)
* Logit ΔROC (train-test): **+0.0241**
* Max ensemble ΔROC:        **+0.1185**

✅ **H1 ПОДТВЕРЖДАЕТСЯ**: ансамбли точнее на тесте, но переобучаются сильнее Log
it.

✅ **H1 ПОДТВЕРЖДАЕТСЯ**: ансамбли точнее на тесте, но переобучаются сильнее Log
it.
```

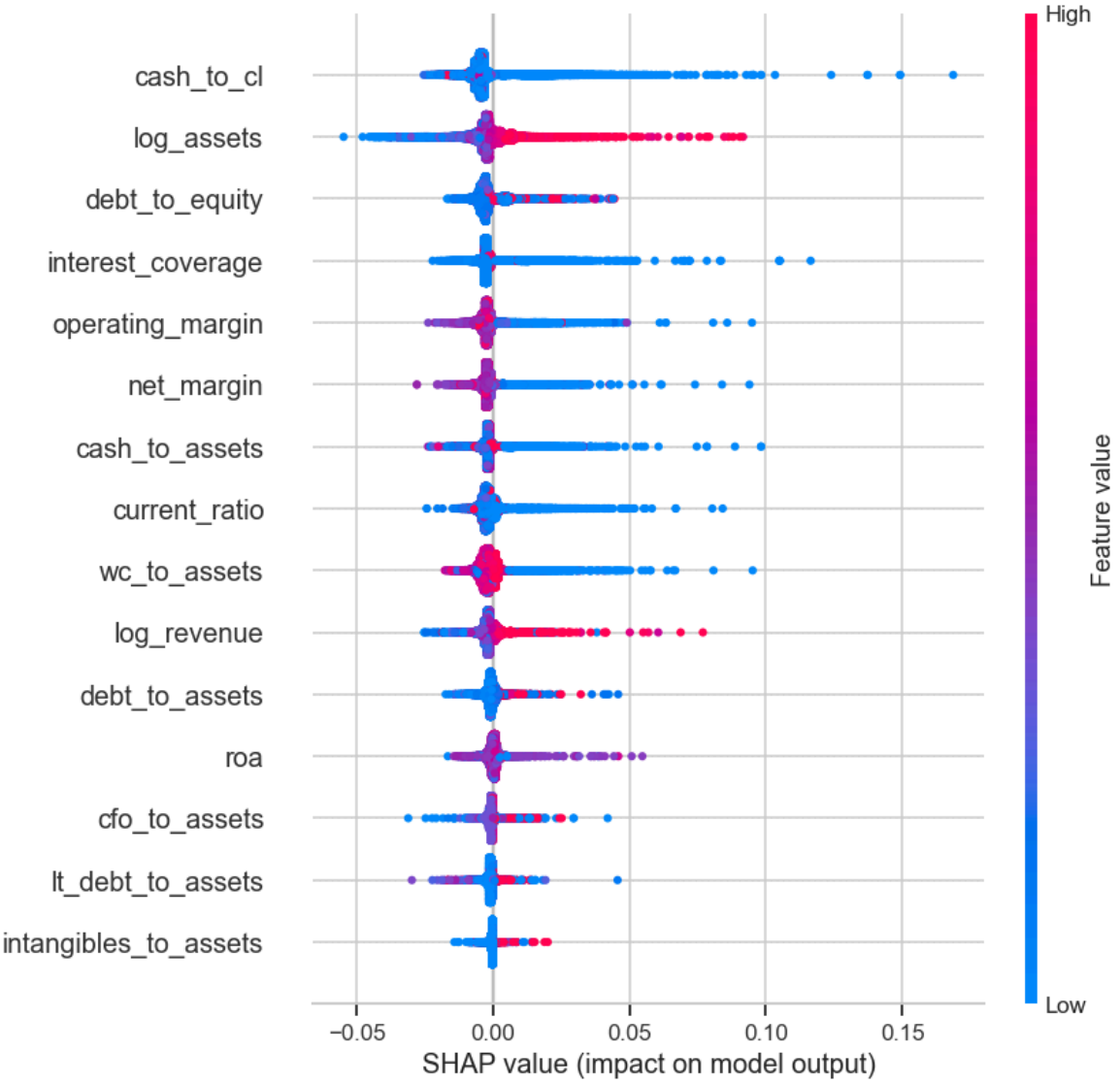
# 6. H2 — SHAP для лучшего ансамбля

In [10]:

```
ensemble_scores = res.loc[['Random Forest', 'XGBoost'], 'ROC-AUC test']
best_name = ensemble_scores.idxmax()
best_model = models[best_name]
print(f'Best ensemble: {best_name}  (test ROC-AUC = {ensemble_scores.max():.4f})

explainer = shap.TreeExplainer(best_model)
sv = explainer.shap_values(X_test)
```

```
# Нормализуем SHAP под бинарную задачу для RF (shap 0.51 может вернуть 3D-массив
if isinstance(sv, list):
    sv = sv[1]
elif hasattr(sv, 'ndim') and sv.ndim == 3:
    sv = sv[:, :, 1]
print('SHAP shape:', sv.shape)

# shap.summary_plot рисует через plt напрямую; ловим фигуру и сохраняем
plt.figure()
shap.summary_plot(sv, X_test, feature_names=FEATURES, show=False)
fig = plt.gcf()
save_fig(fig, '04_shap_summary'); plt.show()
```

```
Best ensemble: Random Forest  (test ROC-AUC = 0.9016)
SHAP shape: (17743, 15)
  saved → ..\reports\russia\04_shap_summary.png
```

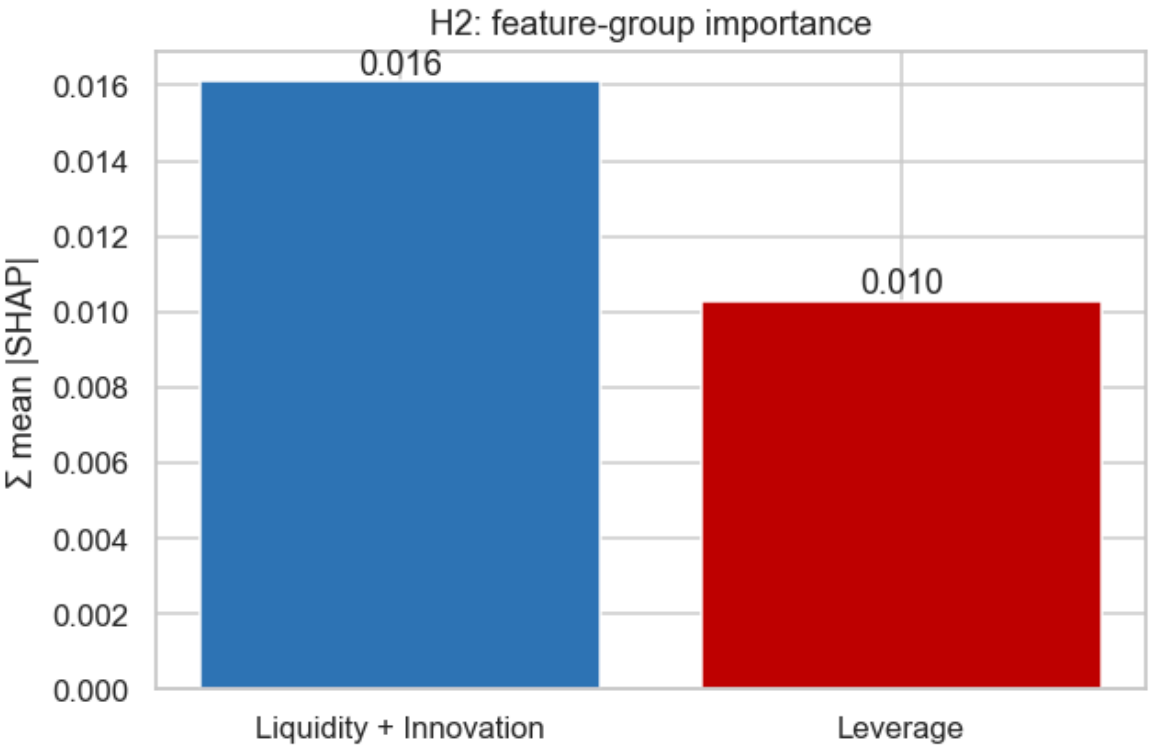



In [11]:

```
mean_abs = np.abs(sv).mean(axis=0)
fi = pd.Series(mean_abs, index=FEATURES).sort_values(ascending=False)
fi.round(4).to_csv(REPORTS / 'ru_shap_feature_importance.csv', header=['mean_abs
print('Top features by mean |SHAP|:')
print(fi.round(4).to_string())

group_sum = {g: fi[cols].sum() for g, cols in FEATURE_GROUPS.items()}
group_df = pd.Series(group_sum).sort_values(ascending=False).round(4)
group_df.to_csv(REPORTS / 'ru_shap_group_importance.csv', header=['sum_abs_shap'
```

```
print('\nΣ|SHAP| по блокам:')
print(group_df.to_string())

liq_inn = group_df.get('Liquidity', 0) + group_df.get('Innovation', 0)
lev     = group_df.get('Leverage',   0)

fig, ax = plt.subplots(figsize=(6, 4))
bars = ax.bar(['Liquidity + Innovation', 'Leverage'],
              [liq_inn, lev], color=['#2E75B6', '#C00000'])
ax.set_ylabel('Σ mean |SHAP|')
ax.set_title('H2: feature-group importance')
for b, v in zip(bars, [liq_inn, lev]):
    ax.text(b.get_x()+b.get_width()/2, v, f'{v:.3f}', ha='center', va='bottom')
plt.tight_layout()
save_fig(fig, '05_h2_group_importance'); plt.show()

log('\n## H2 — Liquidity+Innovation vs Leverage (SHAP)')
log(f'* Best ensemble: **{best_name}** (test ROC-AUC = {ensemble_scores.max():.4
log(f'* Σ|SHAP| Liquidity + Innovation = **{liq_inn:.4f}**')
log(f'* Σ|SHAP| Leverage               = **{lev:.4f}**')
if lev > 0:
    log(f'* Ratio = **{liq_inn/lev:.2f}×**')

if liq_inn > lev:
    verdict2 = '✅ **H2 ПОДТВЕРЖДАЕТСЯ**: в IT-секторе России ликвидность и НМА
else:
    verdict2 = '❌ H2 НЕ подтверждается: рычаг остаётся доминирующим предиктором
log(f'\n{verdict2}')
print('\n' + verdict2)
```

```
Top features by mean |SHAP|:
cash_to_cl               0.0070
log_assets               0.0053
debt_to_equity           0.0040
interest_coverage        0.0037
operating_margin         0.0036
net_margin               0.0030
cash_to_assets           0.0030
current_ratio            0.0029
wc_to_assets             0.0028
log_revenue              0.0025
debt_to_assets           0.0015
roa                      0.0013
cfo_to_assets            0.0011
lt_debt_to_assets        0.0011
intangibles_to_assets    0.0004

Σ|SHAP| по блокам:
Liquidity        0.0157
Leverage         0.0103
Profitability    0.0090
Size             0.0077
Innovation       0.0004
  saved → ..\reports\russia\05_h2_group_importance.png
```




```
## H2 — Liquidity+Innovation vs Leverage (SHAP)
* Best ensemble: **Random Forest** (test ROC-AUC = 0.9016)
* Σ|SHAP| Liquidity + Innovation = **0.0161**
* Σ|SHAP| Leverage               = **0.0103**
* Ratio = **1.56×**

✅ **H2 ПОДТВЕРЖДАЕТСЯ**: в IT-секторе России ликвидность и НМА важнее рычага.
```

✅ **H2 ПОДТВЕРЖДАЕТСЯ**: в IT-секторе России ликвидность и НМА важнее рычага.

## 7. Сохраняем итоговый отчёт

In [12]:
```
summary_path = REPORTS / 'ru_summary.md'
summary_path.write_text('\n'.join(_report_lines), encoding='utf-8')
print(f'✅ Итоговый отчёт: {summary_path}')
print('\nВсе артефакты в reports/russia/:')
for p in sorted(REPORTS.iterdir()):
    print(f'  {p.name}  ({p.stat().st_size/1024:.1f} KB)')
```

```
✅ Итоговый отчёт: ..\reports\russia\ru_summary.md

Все артефакты в reports/russia/:
  01_class_distribution.png  (36.1 KB)
  02_violin_by_class.png  (115.7 KB)
  03_correlation_heatmap.png  (237.6 KB)
  04_shap_summary.png  (166.0 KB)
  05_h2_group_importance.png  (35.6 KB)
  ru_correlation_matrix.csv  (1.8 KB)
  ru_feature_stats.csv  (1.0 KB)
  ru_h1_metrics.csv  (0.2 KB)
  ru_medians_by_class.csv  (0.1 KB)
  ru_shap_feature_importance.csv  (0.3 KB)
  ru_shap_group_importance.csv  (0.1 KB)
  ru_summary.md  (0.8 KB)
```

# 8. Что получилось

В папке `reports/russia/`:

| файл | что внутри |
|---|---|
| `01_class_distribution.png` | столбики баланса классов |
| `02_violin_by_class.png` | violin по 4 ключевым метрикам |
| `03_correlation_heatmap.png` | heatmap корреляций |
| `04_shap_summary.png` | SHAP summary (лучший ансамбль) |
| `05_h2_group_importance.png` | Σ\|SHAP\| по блокам для H2 |
| `ru_h1_metrics.csv` | train/test ROC/PR + Δ по всем трём моделям |
| `ru_shap_feature_importance.csv` | mean \|SHAP\| по фичам |
| `ru_shap_group_importance.csv` | суммы по блокам |
| `ru_correlation_matrix.csv`, `ru_feature_stats.csv`, `ru_medians_by_class.csv` | EDA-таблицы |
| `ru_summary.md` | готовый текст с вердиктами H1 и H2 — можно вставить в §3.2–3.3 |

Всё это — вход для Stage-2 (PIT) и для сравнения с Китаем.